Data & Query Optimization Methods

Data Modeling Methods:

- **Caching Query Results:** stores frequently access data in a high-speed storage layer, such as RAM, to improve performance to frequently accessed data (Snowflake automatically implements data caching, helping increase query performance)

- **Data Compression (lossless compression):** minimizes the volume of digital data, maximizes storage efficiency, and improves data transmission speed, by utilizing:

  - **Pattern Recognition and Encoding:** identifying patterns or sequences and replacing longer character values with shorter representations, think of this like acronyms

  - **Compression Dictionaries:** identify common patterns in the data and uniquely store each pattern in the dictionary and reference it, when necessary, instead of storing every occurrence of the pattern

- **Data Denormalization:**

  - **Pre-Joining Tables:** adding redundant columns from dimension tables into fact tables, reducing the number of joins needed in the query

  - **Table Splitting (Partitioning):**

- **Horizontal Table Splitting:** splitting large tables into multiple smaller tables, each with the same columns, but the data within each table is grouped/split based on certain scenarios, like region or market, year, etc.

- **Vertical Table Splitting:** Splitting a large table into multiple smaller tables, grouping more common attributes together, and less common ones together, applying the same primary key in each table to be able to join the data back to gather when necessary

- **Data Type Optimization:** use appropriate data types to reduce storage and improve query efficiency

- **Indexing:** create indexes on frequently queried columns to speed up data retrieval, especially join columns

- **Materialized Views:** stores the results of pre-computed queries in a separate table, reducing the need for redundant computations, that can then be referenced when needed

- **Online Analytical Processing (OLAP) Environment:** built onto of star and snowflake schemas, that aggregates the data from individual records into commonly used aggregated levels including common attributes and measures. This allows for optimized query performance as the data is already aggregated for increased performance. Can also be related to mirror tables)

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Query Methods:

- **Analyze Query Execution Plans:** use database tools to analyze how queries are executed to identify potential bottlenecks
- **Avoid Subqueries If Possible:** Instead use WITH functions or JOINs
- **Avoid Unnecessary GROUP BY and ORDER BY:** only use these functions when necessary
- **Limit Data Retrieval:**
  - Use WHERE clauses to filter data efficiently, and avoid unnecessary filtering or data manipulation
  - Perform aggregate functions (GROUP BY) before JOINs
  - Avoid using SELECT *, only retrieve columns needed to reduce the amount of data processed
- **Limit The Use of Wildcards:** Using wildcards in LIKFE clauses can lead to full table scans and slower performance